

1. Project Name

Fractal Forge

2. Project Type

Interactive mathematical learning website with visual simulations and a challenge-based learning mode.

3. Project Goal

The goal of Fractal Forge is to help learners understand fractals through explanation, visualization, and interaction.

Fractals can be difficult to understand when they are only explained using definitions and formulas. Many important ideas, such as iteration, self-similarity, fractal dimension, complex numbers, and formula-based behavior, become clearer when they are seen visually.

Fractal Forge transforms these ideas into an interactive website where users can explore how simple mathematical rules create complex structures.

The project focuses on showing that fractals are not just beautiful images. They are mathematical systems created through repeated rules.

4. Math Challenge Addressed

The main mathematical challenge addressed by this project is:

How can abstract fractal concepts be made easier to understand through interactive visualization?

Fractals involve several ideas that are often difficult for learners:

- Iteration
- Self-similarity
- Infinite perimeter and finite area
- Non-integer dimension
- Complex numbers
- Bounded and escaping behavior
- Parameter changes
- Newton's method and root attraction

These concepts are usually taught separately and can feel abstract. Fractal Forge connects them in one learning path and allows users to interact with them directly.

Instead of only reading about a formula, the learner can change values, increase iterations, zoom into structures, and observe how the mathematical behavior changes visually.

5. Project Description

Fractal Forge is an educational website that explains and visualizes fractals step by step.

The website begins with simple geometric fractals, such as the Sierpinski triangle and Koch snowflake. These examples introduce the idea of repeated rules and self-similarity.

Then the project moves into fractal dimension, explaining why some fractals are more complex than a line but do not fully fill a plane.

After that, the website introduces complex numbers and the complex plane. This prepares the learner for formula-based fractals such as Julia sets, the Mandelbrot set, and Newton's fractal.

Each section of the website includes a short explanation and an interactive visual tool.

The project also includes a challenge mode where learners apply the ideas they studied. The challenge mode is not the whole project; it is an active learning component that supports the educational goal.

6. Mathematical Content Included

The project includes the following mathematical topics:

6.1 Fractals and Iteration

A fractal is a structure generated by repeatedly applying a simple rule.

The project introduces this idea using the Sierpinski triangle, where a repeated geometric rule creates a detailed self-similar structure.

6.2 Geometric Fractals

Geometric fractals are created by applying repeated construction rules directly to shapes.

Examples used in the project:

- Sierpinski triangle
- Koch snowflake

6.3 Koch Snowflake

The Koch snowflake is used to show how a simple repeated rule can create an unusual mathematical result.

At every iteration, each side is divided into three parts, the middle part is removed, and it is replaced with two sides of a small outward equilateral triangle.

The number of sides increases according to:

$$N_n = 3 \times 4^n$$

The side length decreases according to:

$$l_n = s / 3^n$$

The perimeter is:

$$P_n = 3s(4/3)^n$$

As n increases, the perimeter approaches infinity, while the area remains finite.

This connects to the coastline paradox.

6.4 Fractal Dimension

The project explains that fractals can have non-integer dimensions.

The self-similarity dimension formula is:

$$D = \log(N) / \log(1/r)$$

For the Koch curve:

$$N = 4$$

$$r = 1/3$$

So:

$$D = \log(4) / \log(3)$$

$$D \approx 1.262$$

This means the Koch curve is more complex than a line but does not fully fill a two-dimensional area.

6.5 Complex Numbers and the Complex Plane

The project introduces complex numbers in the form:

$$z = a + bi$$

Each complex number can be represented as a point on the complex plane.

This allows formula-based fractals to be visualized, because each pixel on the screen can represent a complex number.

6.6 Julia Set

The Julia set is generated using:

$$z(n+1) = z(n)^2 + c$$

In the Julia set, c is fixed and different starting values of z are tested.

The final image is created by coloring points based on whether they stay bounded or escape.

6.7 Mandelbrot Set

The Mandelbrot set uses the same formula:

$$z(n+1) = z(n)^2 + c$$

However, in the Mandelbrot set:

$$z_0 = 0$$

and c changes.

Each point represents a value of c .

The Mandelbrot set shows which values of c create bounded behavior.

6.8 Julia and Mandelbrot Relationship

The project shows that the Mandelbrot set can be understood as a map of Julia sets.

A point selected from the Mandelbrot set can be used as the value of c to generate a corresponding Julia set.

This is one of the main interactive features of the project.

6.9 Newton's Fractal

Newton's fractal is based on Newton's method:

$$z(n+1) = z(n) - f(z(n)) / f'(z(n))$$

The project shows how applying Newton's method across the complex plane creates colored regions based on which root each point approaches.

This connects algebra, calculus, complex numbers, iteration, and visualization.

Working Prototype Draft

1. Prototype Overview

The working prototype of Fractal Forge is an interactive website that demonstrates the main mathematical ideas of the project.

The prototype is designed to show the learning path of the project, from basic geometric fractals to formula-based fractals.

The prototype includes interactive visual tools that allow the user to change values and observe the mathematical results.

2. Prototype Features

Feature 1: Sierpinski Triangle Animation

The prototype includes a Sierpinski triangle animation.

The user can increase the number of iterations.

At each step, the triangle becomes more detailed.

Purpose:

This feature explains how a simple repeated rule creates a fractal.

Feature 2: Koch Snowflake Slider

The prototype includes a Koch snowflake visualizer.

The user can control the iteration level using a slider.

The website displays:

- Number of sides
- Side length
- Perimeter behavior
- Area behavior

Purpose:

This feature explains the idea of infinite perimeter and finite area.

Feature 3: Fractal Dimension Tool

The prototype includes a simple fractal dimension comparison.

The user can compare:

- Line: dimension 1
- Square: dimension 2
- Koch curve: dimension approximately 1.262

Purpose:

This feature explains that fractals can have non-integer dimensions.

Feature 4: Complex Plane Explorer

The prototype includes a complex plane interaction.

When the user moves the cursor over the plane, the website displays the complex number at that location.

Example:

$$z = 0.4 + 0.2i$$

Purpose:

This feature prepares the user for formula-based fractals.

Feature 5: Julia Set Generator

The prototype includes a Julia set generator.

The user can change the real and imaginary parts of c .

When c changes, the Julia set changes.

Purpose:

This feature shows how a fixed parameter controls the appearance of a Julia set.

Feature 6: Mandelbrot Explorer

The prototype includes a Mandelbrot set viewer.

The user can explore the Mandelbrot set and observe its detailed boundary.

Purpose:

This feature shows how changing c creates a map of bounded and escaping behavior.

Feature 7: Mandelbrot-to-Julia Generator

The prototype includes a connection between the Mandelbrot set and Julia sets.

The screen is divided into two parts:

Left side: Mandelbrot set

Right side: Julia set

When the user clicks a point on the Mandelbrot set, that point becomes the value of c , and the corresponding Julia set is generated.

Purpose:

This is the strongest prototype feature because it shows the relationship between the Mandelbrot set and Julia sets directly.

Feature 8: Newton Fractal Visualization

The prototype includes a Newton fractal visualizer.

The user can see colored regions representing different roots.

The user can click a starting point and observe which root it approaches.

Purpose:

This feature explains how Newton's method can create fractal boundaries.

Feature 9: Challenge Mode

The prototype includes a challenge mode.

In this mode, users answer or solve small interactive tasks based on the concepts they learned.

Examples:

- Choose the correct rule for the Koch snowflake
- Classify a fractal as geometric or formula-based
- Adjust c to change a Julia set
- Click a Mandelbrot point to generate a Julia set
- Choose a starting point in Newton's fractal that approaches a specific root

Purpose:

This mode supports active learning by making users apply the concepts.

3. Prototype User Flow

The prototype follows this order:

1. Introduction to fractals
2. Sierpinski triangle interaction
3. Geometric vs formula-based comparison
4. Koch snowflake visualizer
5. Fractal dimension tool
6. Complex plane explorer
7. Julia set generator
8. Mandelbrot explorer
9. Mandelbrot-to-Julia generator
10. Newton fractal visualizer
11. Challenge mode

This order allows the learner to move from simple visual ideas to more advanced formula-based fractals.

4. Prototype Educational Purpose

The prototype is designed to make abstract mathematical ideas visible.

Instead of only reading formulas, the learner can interact with them.

The prototype allows learners to:

- See iteration step by step
- Observe how geometric rules create fractals

- Understand how perimeter can increase without infinite area
 - Compare normal dimension and fractal dimension
 - Use the complex plane visually
 - Change parameters in formula-based fractals
 - Connect the Mandelbrot set with Julia sets
 - See Newton's method as a visual process
 - Apply knowledge through challenges
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Implementation Plan Draft

1. Development Objective

The objective of the implementation plan is to build a working interactive prototype of Fractal Forge before the final submission.

The prototype should clearly demonstrate the mathematical ideas, the interactive tools, and the educational value of the project.

2. Development Stages

Stage 1: Content Preparation

In this stage, the mathematical content is organized and prepared.

Tasks:

- Finalize the explanation script
- Select the main fractals used in the project
- Prepare formulas and definitions
- Decide the order of website sections
- Prepare short explanations for each section

Expected result:

A complete written structure for the website and documentation.

Stage 2: Website Layout

In this stage, the main website structure is built.

Tasks:

- Create the homepage
- Create section layout
- Add navigation between sections
- Design the visual style
- Add placeholders for interactive components

Expected result:

A clear website layout that follows the learning path of the project.

Stage 3: Geometric Fractal Modules

In this stage, the geometric fractal interactions are implemented.

Tasks:

- Build Sierpinski triangle animation
- Build Koch snowflake iteration slider
- Display number of sides
- Display side length behavior
- Display perimeter behavior
- Add fractal dimension comparison

Expected result:

Working geometric fractal visualizations.

Stage 4: Complex Plane Module

In this stage, the complex plane explorer is implemented.

Tasks:

- Create a 2D coordinate plane
- Map screen pixels to complex numbers
- Display the complex number under the cursor
- Prepare the coordinate system for Julia and Mandelbrot rendering

Expected result:

A working complex plane explorer.

Stage 5: Julia Set Module

In this stage, the Julia set generator is implemented.

Tasks:

- Render Julia set using $z(n+1) = z(n)^2 + c$
- Add sliders for the real and imaginary parts of c
- Update the fractal when c changes
- Add escape-time coloring

Expected result:

A working Julia set generator.

Stage 6: Mandelbrot Module

In this stage, the Mandelbrot set explorer is implemented.

Tasks:

- Render Mandelbrot set using $z_0 = 0$ and $z(n+1) = z(n)^2 + c$
- Add zoom or exploration controls
- Color points based on escape behavior
- Allow the user to click a point and read its c value

Expected result:

A working Mandelbrot set explorer.

Stage 7: Mandelbrot-to-Julia Connection

In this stage, the main connection feature is implemented.

Tasks:

- Split the screen into Mandelbrot and Julia panels
- Allow user to click a point in the Mandelbrot set
- Use the clicked point as c
- Generate the corresponding Julia set
- Display the selected c value

Expected result:

A working Mandelbrot-to-Julia generator.

Stage 8: Newton Fractal Module

In this stage, Newton's fractal is implemented.

Tasks:

- Choose a function such as $f(z) = z^3 - 1$
- Calculate roots
- Apply Newton's method to points in the complex plane
- Color regions based on which root each point approaches
- Allow the user to click a point and see the root it approaches

Expected result:

A working Newton fractal visualizer.

Stage 9: Challenge Mode

In this stage, the challenge-based learning mode is added.

Tasks:

- Create concept-based challenge cards
- Add simple scoring or progress tracking
- Connect challenges to website sections
- Include tasks about iteration, Koch snowflake, fractal dimension, Julia sets, Mandelbrot sets, and Newton's fractal

Expected result:

A working challenge mode that supports active learning.

Stage 10: Testing and Final Improvements

In this stage, the prototype is tested and improved.

Tasks:

- Test all interactive features
- Fix visual or calculation errors
- Improve website design
- Make explanations shorter and clearer
- Prepare final demo path
- Prepare documentation for submission

Expected result:

A complete prototype ready for submission and demonstration.

3. Tools and Technologies

The project can be implemented using:

- HTML
- CSS
- JavaScript
- Canvas API
- React if needed
- Mathematical rendering through custom JavaScript functions

The fractal visualizations can be drawn using HTML Canvas because it allows pixel-based rendering, which is suitable for Julia sets, Mandelbrot sets, and Newton's fractal.

4. Implementation Priorities

The project features are divided into must-have and optional features.

Must-Have Features

- Sierpinski triangle animation
- Koch snowflake slider
- Fractal dimension explanation/tool
- Complex plane explorer
- Julia set generator
- Mandelbrot set explorer
- Mandelbrot-to-Julia generator
- Newton fractal visualizer
- Challenge mode

Optional Features

- Custom formula renderer
- Image rendering and export
- Sound interaction with Mandelbrot values
- Custom geometric instruction builder
- AI-assisted fractal rule generation

The must-have features are the focus of the working prototype.

The optional features can be added if time allows.

5. Final Demo Plan

The final demo should show the project in a clear learning sequence.

Demo order:

1. Show the homepage and project goal
2. Show Sierpinski triangle iteration
3. Show Koch snowflake slider and perimeter behavior
4. Show fractal dimension comparison
5. Show complex plane explorer
6. Show Julia set parameter changes
7. Show Mandelbrot set explorer
8. Show Mandelbrot-to-Julia generator
9. Show Newton fractal visualizer
10. Show challenge mode

This demo order shows the full educational journey of Fractal Forge.

Combined Submission Summary

Fractal Forge is an interactive mathematical learning website designed to teach fractals through explanation, visualization, and active learning.

The documentation explains the mathematical background of the project, including iteration, geometric fractals, Koch snowflake, fractal dimension, complex numbers, Julia sets, the Mandelbrot set, Newton's fractal, and related formula-based variations.

The working prototype demonstrates these ideas through interactive tools such as iteration sliders, fractal generators, complex plane exploration, Mandelbrot-to-Julia connection, and Newton fractal visualization.

The implementation plan organizes the project into development stages, beginning with content preparation and website layout, then moving into geometric fractal modules, formula-based fractal modules, challenge mode, testing, and final preparation.

The main purpose of the project is to make fractals easier to understand by turning abstract mathematical behavior into visual and interactive experiences.